Vehicular Edge Computing: A Survey on Architecture and Task Offloading

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Abstract:

With the increasing dependence of communication resources on edge computing, Vehicle Edge Computing (VEC) has become a new mature paradigm in vehicular networks. To fully assist naive readers in clearly understanding the operation logic of VEC, this paper proposes a three-layer architecture of VEC along with some related researchers' study attempts from the perspective of various points of performance. Some common techniques applied in task offloading are also briefly introduced to readers to keep up with the current research progress and also find out some feasible study directions for future to excavate.

Key Words:

Mobile Edge Computing (MEC), Vehicle Edge Computing (VEC), Task Offloading,

I. Introduction

A. Background

The birth of modern communication, coming with the popularization of mobile devices and the exponential eruption of mobile Internet traffic, keeps promoting extraordinary progress in the field of wireless communication and networking. In order to provide computing services for mobile devices at cloud data centers to reach the target of resource maximization sharing, the research field of Mobile Cloud Computing (MCC) is gradually maturing [1] due to its mighty data processing capability. Nevertheless, in the MCC paradigm, the data produced by terminal equipment must experience a long propagation distance to the remote centralized cloud server, which leads to the delay and hence fails to keep pace with the increasingly rigorous requirements of low latency. In the meantime, despite its awesome data processing competence, it is inevitable for MCC to be confronted with pressure generated by the booming number of devices along with their rich applications, which might be dealt with more efficiently, if employing an alternative, better or auxiliary way. Mobile Edge Computing (MEC) appeared to solve this dilemma. On the one hand, MEC can meet the requirements of low latency for the reason that servers are deployed at edge, and operate on terminal devices with faster feedback; on the other hand, MEC sinks content and computing power to provide intelligent traffic scheduling, so that some regional businesses no longer need to spend much time in the cloud to culminate [2]. Feasible to be combined with other existing technologies, MEC has shown excellent performance in many business scenarios, such as Augmented Reality, the Internet of Things, vehicular networks, and so on. Similar to every other communication scenario, a gigantic amount of data needs to be swiftly analyzed and fused for vehicles to make appropriate decisions in vehicular networks, yet the conventional mode such as centralized vehicular cloud computing face huge challenges to meet the requirements in terms of storage and instant communication [3]. For getting rid of this predicament, a new computing mode which is called Vehicle Edge Computing (VEC) has stepped into the field of vision of researchers to make up for the deficiency of centralized processing mode.

B. Motivation and outline of this paper

To actually decentralize computing tasks to the edge of the terminal, the industry

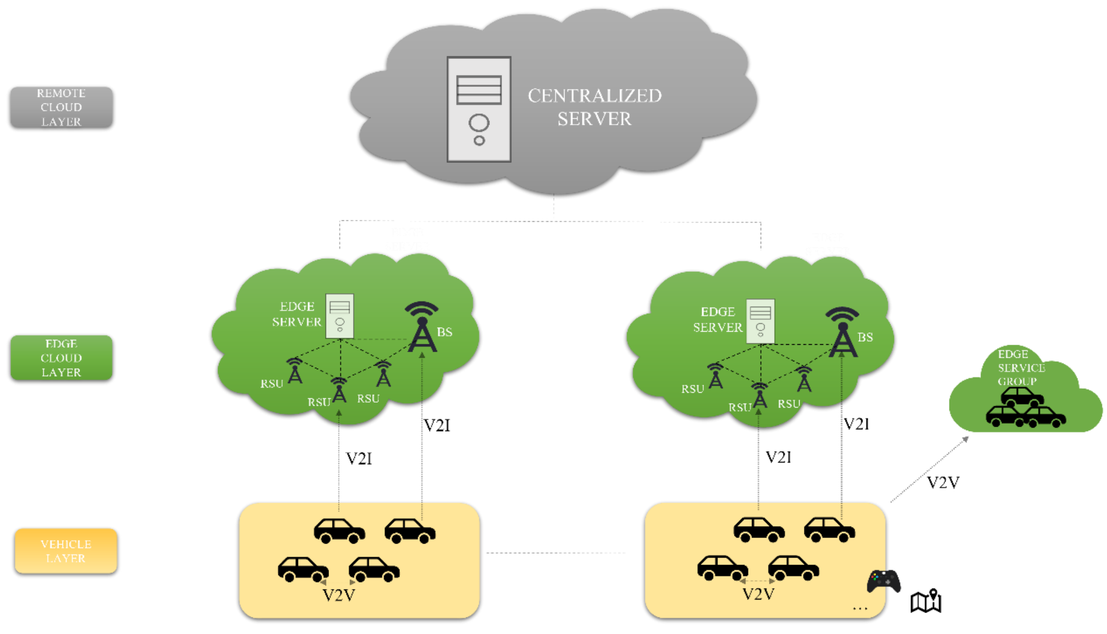

Figure 1: three-layer architecture of VEC

introduced the concept of task offloading in MEC. Task offloading means that the user terminal offloads computing tasks to the MEC network, which mainly aims at ameliorating the shortcomings of equipment in latency, computing performance, energy efficiency, and so on. As the combination of MEC and vehicular networks, VEC also utilizes the key task offloading technique to operate. The writing intention of this paper is to fully figure out the architecture of VEC to assist naive readers in clearly understanding the operation logic of VEC. Besides, although some other crucial techniques play their respective and indispensable roles in VEC, due to the remarkable leading role of task offloading, another purpose of this paper will be displaying or exploring some related researchers' study attempts on the implementation and optimization of task offloading in VEC, so as to keep up with the current research progress and also find out some feasible study directions for future to excavate.

This paper focuses on the architecture of VEC and its core technique: task offloading, which is organized as follows: Section II describes the architecture of VEC and its demands for optimizing performance along with some related work. Section III pays attention to task offloading by elaborating on the common technologies or algorithm models applied in it. Section IV explains the challenges and research potential of VEC; At the end of this paper, the last section summarizes the full text.

II. Architecture of VEC and related work

A. Architecture

The VEC architecture lays its foundation on the following three layers: remote cloud layer, edge cloud layer, and vehicle layer as shown in Figure 1.

(a) Remote cloud layer:

Remote cloud usually has a large scale and powerful computing power, and is often used for large-scale data aggregation, data mining, analysis, optimization and storage of complex data [4]. Computing tasks generated by vehicle applications can be transmitted to remote cloud through the Software Defined Networking, which is a new architecture conducive to dynamic network deployment and agile network management, and the calculation results are then returned to the vehicle. However, since the remote cloud is located at the far end of the network, it might produce quite a lot of time spent to get the calculation results, which leads to unbearable latency in some applications. Therefore, remote cloud is more applicable for those tasks which are not latency-sensitive.

(b) Edge cloud layer:

The edge cloud acts as the intermediary of the communication link between the remote cloud and the vehicles, which serves as an essential component in the structure of VEC. In order to realize the link, the vehicle has devices using wireless communication

Table 1: Comparison between VEC and vehicular cloud computing

Features	Vehicular Edge Computing	Vehicular Cloud Computing
Location	At user’s proximity	Remote location
Latency	Low	High
Communication	Real time	Constraints in Bandwidth
Context awareness	Yes	No
Mobility support	High	Limited
Decision making	Local	Remote
Device Heterogeneity	Highly supported	Limited supported

protocols. The edge cloud can be composed of edge servers, roadside units (RSUs), base stations (BS) and other edge devices with computing, storage and communication capabilities. The base station equipped with MEC server uses vehicle-to-infrastructure (V2I) transmission to collect the calculation tasks of vehicles driving in the coverage area, sends the tasks to MEC server for processing, and finally returns the results to the vehicles, same as the RSUs. In addition, there is an edge service group composed of a number of intelligent vehicles capable of spare computing resources. Both stationary vehicles in the parking lot and vehicles moving on the road can become members of the group and cooperate with each other to complete the offloading task.

Because the edge cloud is closer to the vehicle, the transmission delay from the vehicle to the edge cloud can be greatly reduced, so the edge cloud can provide delay-sensitive services such as low-delay computing, location awareness, real-time video analysis and emergency management for the vehicle. Meanwhile, the edge devices have certain computing power, so the computing pressure of the remote cloud is significantly decreased, and they can cooperate with each other to provide users with higher quality and more reliable services.

(c) Vehicle layer:

The main component of this layer is vehicles; The vehicle in this mode is considered as an intelligent vehicle, and it is completely equipped with many latest sensors and communication equipment such as ultrasonic sensors, onboard wireless device local camera sensor, etc. Vehicles may perform more communication, exchange services and storage [4]. The intelligent vehicle layer contains a group of vehicles with similar geographical locations and shares computing and storage resources through wireless networks to use services from other vehicles as smart vehicular computing, offering the computational and offloading facilities for vehicles available at the edge of the network and providing all the services required by the autonomous vehicles [4].

B. Related work

Compared with traditional vehicular cloud computing, VEC has many outstanding differences due to its unique advantages of edge offloading, which has been methodically displayed in Table 1. Since the optimization of VEC can be analyzed from multiple dimensions from the table, this paper will primarily introduce the research progress of the laboratory and other related researchers from the following three emblematical angles.

(a) Latency

As mentioned above, one of the main purposes of the emergence of no matter MEC or VEC is to solve the delay problem caused by the long-distance cloud computing being too far away from the terminal users, therefore optimizing the latency is the top priority in VEC research. In the following, some work focusing on reducing latency is sorted out.

Authors in [5] focused on the railway communication scene and considered the problem of the shortage of wireless resources on the track side and the channel quality deterioration caused by the high-speed movement of trains, with an eye to solve the urgent problem of using more computing resources to meet the delay demand of latency-sensitive tasks. This paper designed an offloading and caching architecture for rail communication scenarios, and proposed an adaptive computing offloading scheme based on the task preprocessing deep deterministic policy gradient (TP-DDPG) algorithm. Took vehicle users as agents, and learned how to make task offloading and resource allocation decisions according to task status. The convergence and effectiveness of TP-DDPG algorithm were verified by simulation. Compared with other offloading schemes, TP-DDPG algorithm had the advantages of effectively reducing the total delay of the system in the rail transit scenarios in their research work.

As mentioned above, MEC and MCC have their own advantages and MEC can fill the problem of MCC's inevitable delay, but the computing and processing ability is not as powerful as that of the remote cloud computing. Therefore, authors in [6] gave a method of cooperation between them, and also filled the research gap in optimizing both computing offloading strategy and computing resource allocation at the same time. A collaborative computing offloading and resource allocation optimization scheme was proposed, which decoupled the optimization problem into two subproblems, in which game-theoretic approach was used to make the offloading decision. Aiming at this scheme, a distributed computation offloading and resource allocation algorithm was proposed. It iterated between offloading decision and resource allocation, and obtained network elements, optimal computing offloading strategy and computing resource allocation. The numerical results showed that the performance of the scheme was better than the existing methods, which obviously improved the practicability of the system and reduces the task processing delay.

In [7], considering the limited vehicle budget, authors developed an online and distributed edge caching and calculation method to realize delay-sensitive Internet of Vehicle service in an economical and effective way. Performance analysis and a large number of simulation results showed that this method can not only achieve the short-range optimal delay performance within the bounded deviation range, but also flexibly balanced the weighted vehicle running cost and service response delay performance under the time average cost budget constraint.

(b) Energy Efficiency

In order to respond to the call of green energy and create a sustainable, healthy and low-energy-bills vehicular environment, the optimization point of energy efficiency has increasingly entered the field of vision of researchers.

In [8], authors organized a two-stage tandem computation offloading architecture consisting of the mobile devices (MUs)-to- unmanned aerial vehicle (UAV) offloading and UAV-to-BS offloading links for air-ground integrated networks. They studied the computation offloading over the air-ground integrated networks for a scenario where the moving UAV is not only served as an edge server to help the MUs compute their tasks but also as a relay to offload tasks to the BS for processing further. Capturing the time-varying channel conditions and the stochastic traffic models at both the MUs and the UAV, a two-stage tandem queue model is established by combining the heterogeneous offloading processes of the MUs-to-UAV and the UAV-to-BS links. Then, an online distributed mechanism is designed to improve the long-term average EE performance of the mobile entities while trading off the service delay.

(c) Security and privacy

With the severe increase in malicious attacks and eavesdropping, security is an increasingly important issue. In [9], a blockchain empowered group-authentication scheme was proposed for vehicles with decentralized identification based on secret sharing and dynamic proxy mechanism. Sub-authentication results were aggregated for trust management based blockchain to implement collaborative authentication. The edge computing node with a higher-reputation stored in the tamper-proof blockchain could upload the final aggregated authentication result to the central server to achieve the decentralized authentication. This work analyzed typical attacks for this scheme and showed that the proposed scheme achieves cooperative privacy preservation for vehicles while also reducing communication overhead and computation cost.

III. Common models and methods of Task offloading

Task offloading plays a vital role in VEC. Vehicles with limited resources can hand over tasks to other nodes with sufficient resources for computing, so as to speed up processing speed and improve service quality. Task offloading mainly includes two major issues: offloading decision and computing resource allocation, in which offloading decision mainly studies whether and how much mobile terminals need to offload, while computing resource allocation mainly lays stress on where to offload computing tasks that require offloading. To improve the phenomenon of limited computing resources especially when facing dense traffic flow, it is necessary to have a reasonable offloading decision and resource allocation scheme for vehicle computing tasks. The next of this section introduces three types of common technologies in task offloading and their corresponding research progress.

A. Semi-Markov Decision Process (SMDP)

Markov decision process (MDP) is often applied in model decision-making models. In a dynamic system, the state of the system is random, and decisions must be made in each epoch system, and the cost is determined by the decision. However, in many decision-making problems, the time between decision-making stages is not constant, but random [10]. As an extension of MDP, SMDP is used to model stochastic control problems. Different from the MDP, each state of SMDP has a certain stay time, and the stay time is not constant, but random. Therefore, SMDP can be used to model the offloading decision problem. In paper [11], the authors proposed a SMDP-based cloudlet cooperation strategy for the smart mobile device in vehicular networks based on the link duration and the relative mobility characteristics among the buses. The strategy determined offloading/local execution actions based on the state of the smart mobile device to achieve the minimum energy consumption and application completion time.

B. Game Theory

Game theory is well-established as a classic tool to mathematically model the wireless resource allocation problems. Based on the fact that many of the wireless resource allocation problems can be reduced to distributed decision-making problems, game theory becomes an ideal fit. Game theory focuses on strategic interactions among players and thus eliminates the need for a central controller which is a major advantage. In order to determine the offloading of mobile users' data to a specific MEC server, a non-cooperative game was played between mobile users in paper [12], and the unique existence of Nash Equilibrium point was demonstrated. The optimization problem of the user utility function was formulated to determine the optimal offloading data received by the MEC servers and the maximum user utility function. In paper [13], authors studied the power allocation problem for a heterogeneous cellular network. By exploiting interference pricing mechanism, they formulated the interference management problem as a Stackelberg game, and made a joint utility optimization of macrocells and femtocells.

C. Reinforcement Learning (RL)

Reinforcement learning is another method for distributed decision-making. In RL, autonomous agents learn the best behavior through rewards and punishments obtained in each round of games. Because agents don't know which operation is the best, they learn by balancing the exploration of unknown operations and the exploitation of existing knowledge. That is to say, Agent learns by "trial and error", and the reward obtained by interacting with the environment guides the behavior, with the goal of making the agent get the maximum reward [10]. Some common RL methods include Multi-armed Bandit, Q-learning, etc. In addition, Deep Reinforcement Learning, (DRL) can be used to solve the resource management problem. At present, there are some vehicles that use DRL to make offloading decisions and allocate resources. Paper [14] used the DRL method based on incentive mechanism to reduce implementation complexity in vehicular fog computing, where offloading decisions could be generated faster through deep neural network. Authors in paper [15] proposed a two-phase solution based on the federated learning to solve the nonconvex problem in the paradigm of platooning unmanned ground vehicles.

IV. Challenges and research directions

A. Vehicle Mobility

Different from the traditional communication network, the vehicular network is highly dynamic and unstable because of the high-speed movement of vehicles and their unpredictable real-time change of state, which also brings great challenges to the task offloading in the vehicle edge network. First of all, the movement of vehicles will lead to changes in the nearby communication environment, which will cause the communication link to receive large and small interference, thus causing changes in data transmission rate and power consumption, and the communication quality will decline, which will bring latency. Secondly, the high-speed moving vehicles will cause the edge nodes to move at a high speed, and the communication link duration will be very short, resulting in frequent links and disconnections, which will lead to the failure of the vehicle-to-everything communication and calculation task transmission. In addition, if the vehicle leaves the communication range while performing task offloading or waiting for task offloading, the calculation of offloading tasks will also fail. Vehicle mobility and its influence are inevitable, but it can be alleviated through practice and attempt. The research direction can be to make use of the small speed difference of vehicles traveling in the same direction, so that the vehicles traveling in the same direction can be used as the candidate set for task offloading. It is also possible to sacrifice some communication expenses and network resources appropriately to make up for the problem that the vehicle is easy to leave the communication range and the task fails. The compromise between failure probability and cost of communication resources is the key point worthy of analysis and research.

B. Strict low latency constraints

With the continuous improvement of users' requirements for the service quality of communication networks, the user experience index of vehicle-mounted networks has also become a noticeable point for network designers and maintainers. For users who use smart cars, vehicles have rich application functions and the corresponding delays of various applications are indispensable. The complex functions of in-vehicle applications produce a large amount of data that needs to be processed, and it is inevitable that the precise and complete transmission of these data will conflict with the strict low delay limit. Of course, the changeable vehicle network topology mentioned above will also bring unwanted delay. At the same time, delay counts a great deal in some vehicle applications that decide the driver's safety conditions, such as automatic early warning system and emergency information broadcast. Subsequent research on reducing delay can be further optimized from two major directions: efficiency-oriented and safety-oriented.

C. Algorithm complexity of task offloading

Offloading is an attractive solution to meet the demand of vehicle applications for response time. In the offloading process, besides the normal computing and communication time, the time complexity of decision algorithm will affect the overall response time of the application. Time complexity is an important indicator to measure the quality of decision algorithm. The higher the time complexity, the more time is needed for decision-making, which is not suitable for offloading computing tasks with high real-time requirements. Based on DRL, the problem of high real-time computation offloading may be solved. DRL has a strong ability to deal with complex problems and can effectively learn from experience, so it can obtain better offloading strategies in various complex situations of VEC, which provides a good idea for the decision-making of computation offloading of complex models. VEC computation offloading based on DRL could be the next hot issue of research [10].

V. Conclusion

To conclude, this paper has introduced the background and concepts of VEC and then discussed the architecture for VEC. Task offloading plays an essential part in the VEC system, which is thought of a suitable technique for vehicles to reduce latency, increase EE and security performance. Although VEC has a great advantage in make up for on-board computing resources and shorten the transmission route compared with that of the remote cloud, it still has a long way to go and is faced with many tricky challenges to tackle and optimize. So, in the last of this essay, it concludes some challenges and puts forward a little possible research directions for academics and researchers related to this field.

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